

Chemical Bonding

Ionic and Covalent Bonds

CHEMISTRY · CLASS 8 · 3-LESSON UNIT · ~40–45 MIN EACH

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Unit Overview

This unit introduces students to why and how atoms combine to form chemical bonds, focusing on the two principal bond types relevant to the grade 8 syllabus: ionic (electrovalent) bonding and covalent bonding. Across three connected lessons, students progress from the concept of valence electrons and stability, through the electron-transfer model of ionic bonding, to the electron-sharing model of covalent bonding, before consolidating both models and relating bond type to observable physical properties.

Unit Learning Objectives

- Explain why atoms form chemical bonds in terms of achieving a stable (noble gas) electronic configuration.
- Describe, using diagrams, the formation of ionic bonds by transfer of electrons between metal and non-metal atoms.
- Describe, using diagrams, the formation of covalent bonds by sharing of electron pairs between non-metal atoms.
- Draw dot-and-cross diagrams for simple ionic compounds (e.g. NaCl, MgO) and covalent molecules (e.g. Cl₂, H₂O, CH₄).
- Relate the type of bonding present in a substance to its physical properties (melting/boiling point, electrical conductivity, solubility).
- Distinguish between ionic and covalent compounds using structure and property-based evidence.

Prior Knowledge Required

- Basic atomic structure: protons, neutrons, electrons, and electron shell arrangement.
- Periodic table groups and the link between group number and number of valence electrons.
- The concept of ions (cations and anions) and how they are formed.

Key Vocabulary

Valence electron, octet rule, ion (cation / anion), ionic bond, electrovalent bond, covalent bond, dot-and-cross diagram, lattice structure, molecule, electronegativity (introductory level).

Resources & Materials

- Whiteboard/marker, periodic table charts (one per student or shared).
- Molecular model kits or coloured beads/plasticine and cocktail sticks for model-building.
- Printed worksheets: dot-and-cross diagram practice sheets (Lessons 1 and 2).
- Samples or images: table salt (NaCl), sugar, a length of copper wire — for the properties comparison activity.
- Projector/slides summarising electron configurations of common elements (Z = 1–12).

Assessment Strategy (across the unit)

- **Formative:** whiteboard mini-checks, questioning during modelling, worksheet monitoring, exit tickets at the end of each lesson.

- **Summative:** end-of-unit quiz (Lesson 3) covering diagram drawing, bond identification, and property prediction.

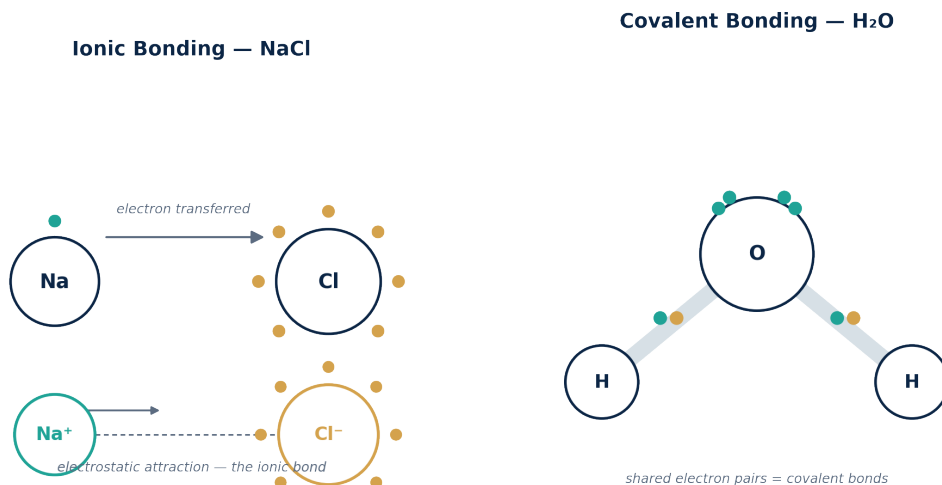


Figure 1. Dot-and-cross diagrams — electron transfer forms the ionic bond in NaCl; electron sharing forms the covalent bonds in H₂O.

Introduction to Bonding & Ionic Bonds

Learning Objectives

- State the octet rule and explain why atoms bond in terms of achieving stability.
- Describe the formation of an ionic bond as the transfer of electrons from a metal to a non-metal atom.
- Draw a simple dot-and-cross diagram for NaCl.

Stage	Time	Teacher Activity	Student Activity
Starter	5 min	Show images of table salt and a metal wire. Ask: "What holds the atoms together in these substances?" Elicit prior ideas.	Discuss in pairs and share initial ideas with the class.
Introduction	10 min	Recap electron shell diagrams for Na and Cl. Introduce the octet rule and noble gas stability using Ne and Ar as reference points.	Draw electron shell diagrams for Na and Cl in notebooks; identify valence electrons.
Modelling	15 min	Model the formation of NaCl step by step on the board: electron transfer from Na to Cl, resulting Na^+ and Cl^- ions, and electrostatic attraction forming the ionic bond. Introduce dot-and-cross notation.	Copy the modelled diagram, labelling ion charges and the electron transferred.
Guided Practice	10 min	Circulate while students attempt MgO and CaCl_2 dot-and-cross diagrams on the worksheet; provide targeted feedback.	Complete worksheet diagrams for MgO and CaCl_2 individually, checking with a partner.
Plenary	5 min	Ask 2–3 students to explain their MgO diagram aloud. Issue an exit ticket: "Why does Na form Na^+ and not Na^{2+} ?"	Answer the exit-ticket question individually before leaving.

Support	Extension
Provide partially completed dot-and-cross diagrams for students to finish.	Ask higher-attaining students to attempt Al_2O_3 and explain the need for balancing charges.

Homework: Complete dot-and-cross diagrams for KCl and MgCl_2 , with one sentence explaining each ion's charge.

Covalent Bonding

Learning Objectives

- Explain covalent bonding as the sharing of electron pairs between non-metal atoms.
- Draw dot-and-cross diagrams for simple and multiple covalent bonds (Cl_2 , H_2O , O_2 , CH_4).
- Distinguish single, double, and (introductory) triple covalent bonds.

Stage	Time	Teacher Activity	Student Activity
Starter	5 min	Quick recall quiz on ionic bonding from Lesson 1 (mini-whiteboards): 3 rapid-fire questions.	Answer on mini-whiteboards and hold up for instant teacher feedback.

Stage	Time	Teacher Activity	Student Activity
Introduction	8 min	Pose the question: "What happens when two non-metals need electrons — can either one give them away?" Introduce electron sharing as the alternative route to stability.	Discuss why transfer doesn't work between two non-metals.
Modelling	15 min	Model Cl_2 (single bond) and O_2 (double bond) formation using dot-and-cross diagrams; use molecular model kits/beads to build 3D representations of H_2O and CH_4 .	Build their own H_2O and CH_4 models using kits/beads, then sketch the corresponding dot-and-cross diagrams.
Guided Practice	12 min	Support students working through a worksheet covering N_2 , HCl , and CO_2 ; question groups on shared pairs vs. lone pairs.	Complete worksheet diagrams in pairs, identifying number of shared electron pairs in each molecule.
Plenary	5 min	Cold-call students to state one similarity and one difference between ionic and covalent bond formation.	Respond verbally; note the comparison in notebooks.

Support	Extension
Focus on single-bond molecules only (Cl_2 , H_2O) with a structured writing frame.	Introduce N_2 as a triple-bond example and ask students to justify the number of shared pairs.

Homework: Draw dot-and-cross diagrams for NH_3 and C_2H_6 (or CH_3OH for extension), labelling shared and lone pairs.

LESSON 3

40–45 min

Comparing Bonds, Properties & Assessment

Learning Objectives

- Compare ionic and covalent bonding in terms of formation, structure, and resulting properties.
- Predict physical properties (melting point, conductivity, solubility) from bond type.
- Demonstrate unit learning through a short summative quiz.

Stage	Time	Teacher Activity	Student Activity
Starter	5 min	Display samples/images of NaCl and sugar (or wax). Ask students to predict which is ionic and which is covalent, and why.	Make and justify predictions in notebooks based on prior lessons.
Practical / Demo	15 min	Conduct or demonstrate a simple conductivity test (solid vs. dissolved-in-water NaCl) and compare melting behaviour of an ionic vs. a covalent solid. Guide students to link results to structure (giant ionic lattice vs. simple molecular structure).	Record observations in a results table; complete guided questions linking results to bonding type.
Consolidation	10 min	Build a comparison table on the board (formation, particles involved, melting/boiling point, conductivity, solubility) with class input.	Copy and complete the comparison table in notebooks.
Summative Quiz	12 min	Administer a short written quiz: 2 dot-and-cross diagrams (1 ionic, 1 covalent), 3 property-prediction questions, 1 short-answer comparison question.	Complete the quiz individually under timed conditions.

Stage	Time	Teacher Activity	Student Activity
Plenary	3 min	Review quiz answers briefly as a class; preview the next topic (metallic bonding or giant covalent structures, per syllabus sequence).	Self-check answers against the model responses shown.

Support	Extension
Provide the comparison table with headings pre-filled, requiring students to complete content only.	Ask students to explain, in writing, why ionic compounds conduct electricity only when molten or in solution, but not as solids.

Homework: Past-paper style questions on ionic and covalent bonding (as available from PISR/PISK or O-Level Chemistry syllabus resources), reviewed at the start of the next unit.

Overall Unit Assessment Criteria

Knowledge & Understanding	Correctly explains octet rule and distinguishes ionic vs. covalent bond formation
Application (Diagrams)	Accurately draws dot-and-cross diagrams for given compounds, including charges/shared pairs
Analysis	Links bonding type to physical properties with correct reasoning
Communication	Uses correct chemical terminology (ion, valence electron, lattice, molecule) consistently